

NICHOLAS A LOMBARDO, PHD

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Education

2025	Ph.D. Earth and Planetary Sciences	Yale University
	<i>Dissertation: "The Dynamics and Composition of the Middle Atmosphere of Titan"</i>	
2021	M.Phil. Earth and Planetary Sciences	Yale University
2017	B.S. Physics & Mathematics	Central Conn. State Univ.

Additional Training

2024	Mission Design School: Planetary Science (PSSS; PI)	10 wk	Caltech Jet Propulsion Lab
2024	Tracer Mixing in Fluids Summer School	2 wk	Brin Ctr of Mathematics, UMD

Professional Appointments

2026-present	Consultant	NASA Goddard Space Flight Center under SURA
2026-present	Lecturer	University of Hartford Dept. of Physics
2026-present	Visiting Lecturer	Trinity College Dept. of Physics
2023-present	Lecturer	Central Connecticut State Univ. Dept. of Physics
2019–2025	Research & Teaching Fellow	Yale University Dept. of Earth & Planetary Sciences
2017–2019	Research Assistant	NASA Goddard Space Flight Center through UMBC
2016–2017	Intern	NASA Goddard Space Flight Center
2013–2016	STEM Tutor	Central Connecticut State Univ. TRiO Office

Space Mission Involvement

2025–present	Titan Science Case Dev. Lead	<i>Habitable Worlds Observatory</i>
2019–present	Student Associate	<i>Dragonfly</i> Mission to Titan (NASA New Frontiers 4)
2016–2019	Research Assistant	<i>Cassini</i> Composite Infrared Spectrometer Team

Honors and Awards (underline – financial awards)

<u>2024</u>	Yale Estwing Hammer Prize – <i>Outstanding Graduate Student in Earth and Planetary Sciences</i>
<u>2019</u>	Yale Geology and Geophysics Bateman Fellowship
2018	NASA Group Achievement Award – <i>Cassini CIRS Investigation Team for Grand Finale Orbits</i>
2018	NASA Group Achievement Award – <i>Cassini CIRS Team for Solstice, F-Ring, and Proximal Orbits</i>
2017	CCSU Physics and Engineering Physics Departmental Honors
2016	CCSU John Bulman Award – <i>Service to the Department of Physics and Engineering Physics</i>
<u>2013</u>	CCSU-NSF Computer Science, Math, and Physics Scholarship (4 years)

University Teaching

Instructor of Record

Spring 2026	17 students	Trinity College	Phys 231 Electricity and Magnetism Laboratory
"	30 students	University of Hartford	Phy 102 Electricity and the Human Body
"	25 students	"	Phy 102 Electricity and the Human Body
"	30 students	"	Phy 122 Physics for Architecture
"	17 students	Central CT State Univ.	Phys 121 General Physics I Laboratory
"	34 students	"	Phys 121 General Physics I
"	17 students	"	Phys 126 University Physics II Laboratory
Fall 2024	17 students	"	Phys 126 University Physics II
Spring 2024	16 students	"	Phys 126 University Physics II Laboratory
Spring 2023	17 students	"	Phys 122 General Physics II Laboratory

Graduate Teaching Fellow

Spring 2025	13 students	Yale Univ. ENV 774 Framing Climate Migration Science/Policy/Ethics
Fall 2023	120 students	“ EPS 274 Fossil Fuels and World Energy
Fall 2019	27 students	“ G&G 140 Intro. to Atmosphere, Ocean, and Climate Change

Completed Grants

Research Grants

- 2024 NSF Astro.: *Investigating the Chemical and Dynamical Complexity of Titan's Upper Atmosphere* (Collaborator)
- 2024 NASA Solar System Observations: *Winds of change: a multi-decade study of Titan's middle atmosphere across seasons* (Collaborator)

Awarded Observing Proposals

- 2024 7 hr Atacama Large Mm/sub-mm Array: *Winds of Change: Temporal Variability of Titan's High-Altitude Circulation System* (Co-I; PI: Cordiner)
- 2023 12 hr Hale Telescope, Palomar Observatory: *Characterizing Titan's stratospheric wind profile via adaptive optics imaging of a stellar occultation* (Collaborator; PI: Marlin)
- 2022 3 hr Faulkes Telescope N., Las Cumbres Observatory: *Constraining Titan's Stratospheric Winds near Equinox via a Stellar Occultation* (Principal Investigator)
- 2022 12 hr NASA Infrared Telescope Facility: *Investigating Nitrile Chemistry on Titan* (Co-I; PI: Coustenis)

Travel Grants

- 2025 AAS Division for Planetary Sciences Hartmann Student Travel Grant
- 2025 Caltech Earth and Planetary Clouds Workshop Early-Career Travel Grant
- 2022 Europlanet Science Congress Student Travel Grant
- 2021 Titan Through Time V Student Travel Grant
- 2019 AAS Division for Planetary Sciences Hartmann Student Travel Grant
- 2016 CCSU Faculty Student Research Grant & CCSU Alumni Travel Grant

Refereed Publications

- in rev. **Lombardo, N. A.** & Lora, J. M., “Interaction of Trace Species and Three-Dimensional Circulation in Titan's Middle Atmosphere”, *under review in JGR: Planets*
2025. Shultis, J., Waugh, D. W., Toigo, A. D., **Lombardo, N. A.**, Lora, J. M., “Impact of Titan's Polar Vortex on the Transport of Chemical Species”, *Journal of Geophysical Research: Planets*, **131**, 2
2025. Marlin, T. C., Young, E. F., de Kleer, K., Cordiner, M., **Lombardo, N. A.**, de Pater, I., Lora, J. M., Corlies, P., Cosentino, R., Nixon, C. A., Rodriguez, S., Thelen, A., “Zonal Winds in Titan's Middle Atmosphere from a Stellar Occultation Observed with Keck Adaptive Optics”, **6**, 286
2025. Wright, L., Teanby, N. A., Irwin, P. G. J., Nixon, C. A., **Lombardo, N. A.**, Mitchell, D., Lora, J. M. “Seasonal Evolution of Titan's Stratospheric Tilt and High-Resolution Temperatures, Zonal Winds, and Polar Vortices”, *The Planetary Science Journal*, **6**, 114
2025. Nixon, C. A., Bézard, B.†, Cornet, T.†, Coy, B. P.†, de Pater, I.†, Es-Sayeh, M.†, Hammel, H. B.†, Lellouch, E.†, **Lombardo, N. A.**†, López-Puertas, M.†, Lora, J. M.†, Rannou, P.†, Rodriguez, S.†, Teanby, N. A.†, Turtle, E. P.†, +30 others, “Titan's Atmosphere in Late Northern Summer from JWST and Keck Observations”, *Nature Astronomy*, †*These authors contributed equally*
2024. Bézard, B., Nixon, C. A., Vinatier, S., Lellouch, E., Greathouse, T. K., Jiles, R., **Lombardo, N. A.**, Jolly, A., “The D/H ratio in Titan's acetylene from high spectral resolution IRTF/TEXES observations”, *Icarus*, **421**, 116221
2024. Light, S., Gurwell, M., Thelen, A., **Lombardo, N. A.**, Nixon, C. A., “Measurements of Titan's Stratospheric Winds During Equinox with the eSMA”, *The Planetary Science Journal*, **5**, 98

2023. **Lombardo, N. A.** & Lora, J. M., “The Heat and Momentum Budgets of Titan’s Middle Atmosphere”, *Journal of Geophysical Research: Planets*, **128**, e2023JE00806
2023. Lewis, N. T., **Lombardo, N. A.**, Read, P. T., Lora, J. M., “Equatorial Waves and Superrotation in the Stratosphere of a Titan General Circulation Model”, *The Planetary Science Journal*, **4**, 149
2023. Coy, B. P., Nixon, C. A., Rowe-Gurney, N., Achterberg, R. K., **Lombardo, N. A.**, Fletcher, L. N., Irwin, P. G. J. “Spitzer IRS Observations of Titan as a Precursor to JWST MIRI Observations”, *The Planetary Science Journal*, **4**, 114
2023. **Lombardo, N. A.** & Lora, J. M., “Influence of Observed Seasonally Varying Composition on Titan’s Stratospheric Circulation”, *Icarus*, **390**, 115291
2022. Steffans, B.L., Nixon, C.A, Sung, K., Irwin, P.G.J., **Lombardo, N.A.**, Pereira, E. “New Constraints on Titan’s Stratospheric n-Butane Abundance”, *The Planetary Science Journal*, **3**, 59
2020. Hewett, D., Bernath, P. F., Wong, A., Billingham, B. E., Zhao, J., **Lombardo, N. A.**, Nixon, C. A., Jennings, D. E. “N₂ and H₂ Broadened Isobutane Infrared Absorption Cross Sections and Butane Upper Limits on Titan”, *Icarus*, **344**, 113460
2019. Nixon, C. A., Ansty, T. M., **Lombardo, N. A.**, Bjoraker, G. L., Achterberg, R. K., Annex, A. M., Rice, M., Romani, P. N., Jennings, D. E., Coustenis, A., Bézard, B., Vinatier, S., Lellouch, E., Teanby, N. A., Cottini, V., Flasar, F. M. “Cassini Composite Infrared Spectrometer Observations of Titan 2004-2017”, *The Astrophysical Journal Supplement Series*, **244**, 14
2019. ***Lombardo, N A.**, Nixon, C. A., Greathouse, T. K., Bézard, B., Jolly, A., Vinatier, S., Teanby, N. A., Richter, M. J., Irwin, P. J. G., Coustenis, A., Flasar, F. M. “Detection of Propadiene on Titan”, *The Astrophysical Journal Letters*, **881**, L33 - *featured on Sci-News and NASA Astrobiology Institute homepage
2019. **Lombardo, N. A.**, Nixon, C. A., Sylvestre, M., Jennings, D. E., Teanby, N., Irwin, P. G. J., and Flasar, F. M. “Ethane in Titan’s Stratosphere from Cassini CIRS Far- and Mid-Infrared Spectra”, *The Astronomical Journal*, **157**, 160-170
2019. **Lombardo, N. A.**, Nixon, C. A., Achterberg, R. K., Sung, K., Jolly, A., Irwin, P. G. J., and Flasar, F. M. “Spatial and Seasonal Variations in C₃H₈ Hydrocarbon Abundance in Titan’s Stratosphere with Cassini CIRS Observations”, *Icarus*, **317**, 454 – 469

Professional Service

University Service

- 2021–2024 Student representative on Yale Earth and Planetary Sciences Colloquium Committee
- 2020–2021 Undergraduate & Graduate Student Mentor, Yale Dept. of Earth and Planetary Sciences
- 2014–2016 President of the Central CT State Univ. Society of Physics Students

Conference Service

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| 2025 | Co-convener: Exploration of Titan; | EPSC-DPS Joint Meeting |
| 2024 | Virtual Session Chair: Io; | AAS Division for Planetary Sciences 56 th Meeting |
| 2024 | Session Chair: Venus; | AAS Division for Planetary Sciences 56 th Meeting |
| 2024 | Science Organizing Committee member; | AAS Division for Planetary Sciences 56 th Meeting |
| 2017 | Poster Session Lead; | Titan Through Time IV Workshop |

Journal Referee

The Astronomical Journal, The Astrophysical Journal, Geophysical Research Letters, New Astronomy, The Planetary Science Journal, Yale Undergraduate Research Journal

Proposal Review Panels

Panelist, secretary, and external reviewer for NASA Planetary Science

Invited Colloquia and Seminars

- 2025 School of Earth Sciences Seminar, University of Bristol
- 2025 Planetary Science Colloquium, Southwest Research Institute, Boulder, CO
- 2025 Yuk Lunch Seminar, Department of Geology and Planetary Science, Caltech
- 2025 Departments of Physics & Astronomy and Earth & Climate Science Joint Seminar, Bates College
- 2024 Department of Earth and Planetary Sciences Seminar, Johns Hopkins University
- 2023 Department of Mathematics Colloquium, Central Connecticut State University
- 2022 Department of Geological Sciences Seminar, Central Connecticut State University
- 2021 Department of Physics and Engineering Physics Seminar, Central Connecticut State University
- 2019 School of Physics and Astronomy Space Physics Seminar, University of Leicester
- 2018 Center for Research in Exploration in Space Science and Tech. Seminar, NASA GSFC
- 2018 Department of Geological Sciences Seminar, Central Connecticut State University

First-Authored Conference and Workshop Talks

- 2025. **Lombardo, N. A.***, de Batz de Trenqueleon*, B., Shultis, J., Lora, J. M., Rannou, P., Waugh, D., Lian, Y., Newman, C. “The Titan Middle Atmosphere Intercomparison Project”, Europlanet Science Congress – Division for Planetary Sciences Joint Meeting **--these authors contributed equally*
- 2025. **Lombardo, N. A.** & Lora, J. M., “Simulating Clouds in the Middle Atmosphere of Titan”, Caltech Earth and Planetary Clouds Workshop
- 2024. **Lombardo, N. A.** & Lora, J. M. “Feedbacks between Idealized Chemical Tracers and Dynamics through Radiative Heating in Titan’s Middle Atmosphere”, 55th Division for Planetary Sciences Meeting
- 2023. **Lombardo, N. A.**, et al., “Measurement of Hydrogen Isocyanide in Titan’s Lower Stratosphere”, American Geophysical Union
- 2023. **Lombardo, N. A.** & Lora, J.M. “Simulating the Production of Stratospheric Ice Clouds over Titan’s Winter Pole with the Titan Atmospheric Model”, 54th Division for Planetary Sciences Meeting
- 2023. **Lombardo, N. A.** & Lora, J. M. “Titan’s Stratospheric Heat and Momentum Budgets”, Titan Through Time VI
- 2022. **Lombardo, N. A.** & Lora, J.M. “Interaction of Chemical Tracers with Titan’s General Circulation”, American Geophysical Union
- 2022. **Lombardo, N. A.** & Lora, J. M., “The Energy and Momentum Budgets of Titan’s Stratospheric Polar Vortex”, 53rd Division for Planetary Sciences Meeting
- 2022. **Lombardo, N. A.** & Lora, J. M., “The Energy and Momentum Balance of Titan’s Stratospheric Polar Vortex as Simulated in a General Circulation Model”, Europlanet Science Congress Meeting
- 2019. **Lombardo, N. A.**, et al., “Detection of Allene on Titan,” Europlanet Science Congress – Division for Planetary Sciences Joint Meeting
- 2019. **Lombardo, N. A.**, et al., “Detection of Allene on Titan,” Hydrocarbon Worlds NAI Workshop
- 2018. **Lombardo, N. A.**, et al., “Using Far-Infrared Observations to Probe Ethane Abundance in Titan’s Lower Stratosphere,” 50th Division for Planetary Sciences Meeting
- 2018. **Lombardo, N. A.**, et al. “C₃H_X Hydrocarbon Abundance in Titan’s Atmosphere from *Cassini* Infrared Spectra,” 49th Lunar and Planetary Science Conference
- 2017. **Lombardo, N. A.**, et al. “Abundance Profiles for C₃ Hydrocarbons in Titan’s Atmosphere,” 49th Division for Planetary Sciences Meeting
- 2015. **Lombardo, N. A.**, et al., “Origin of Photoluminescence in Graphene Oxide”, APS NES Fall

Select First-Authored Conference and Workshop Posters

2025. **Lombardo, N. A.**, Nixon, C. A., Koskinen, T. T., “Exploring Titan Dynamic Atmosphere with the Habitable Worlds Observatory”, Towards the Habitable Worlds Observatory: Visionary Science and Transformative Technology
2024. **Lombardo, N. A.** & Lora, J. M., “Seasonal-Scale Transport of Heat and Momentum in Titan’s Middle Atmosphere”, 24th Conference on Atmospheric & Oceanic Fluid Dynamics
2024. **Lombardo, N. A.** & Lora, J. M. “Interaction of Trace Species with Titan’s General Circulation”, 24th Conference on Atmospheric & Oceanic Fluid Dynamics
2023. **Lombardo, N. A.**, et al., “Measurement of Hydrogen Isocyanide (HNC) in Titan’s Lower Stratosphere”, Titan Through Time VI
2022. **Lombardo, N. A.**, et al., “Variability and Trends in Beaufort Gyre Near-Inertial Wave Activity inferred from Bottom Pressure Measurements, 2003—2021”, Ocean Sciences Meeting
2021. **Lombardo, N. A.** & Lora, J.M., “Dynamical Simulations of Titan’s Stratosphere using Observationally Derived Molecular Abundance and Aerosol Opacity,” Titan Through Time V
2019. **Lombardo, N. A.**, et al., “Detection of Allene in Titan’s Atmosphere,” Astrobiology Science Conf.
2018. **Lombardo, N. A.**, et al., “The Search for Allene in Titan's Stratosphere using Mid-IR Spectra from the TEXES Instrument on IRTF,” American Geophysical Union, Fall Meeting
2018. **Lombardo, N. A.**, et al., “Comparison of Modeled Alkane Abundances in Titan’s Atmosphere using Different Spectral Windows with *Cassini* CIRS spectra,” Cassini Science Symposium
2018. **Lombardo, N. A.**, et al., “C3 Hydrocarbon Abundance in Titan's Atmosphere with *Cassini* Infrared Spectra”, American Astronomical Society, AAS Meeting #231
2017. **Lombardo, N. A.**, et al., “A Sensitive Search for Allene on Titan”, Titan Through Time IV
2016. **Lombardo, N. A.**, et al., “Synthesis and Characterization of $\text{CoFe}_{2-x}\text{Ni}_x\text{O}_4$ Magnetic Nanoparticles”, APS NES Spring
2016. **Lombardo, N. A.**, et al., “Controllably Inducing and Modeling Optical Response from Graphene Oxide,” APS March Meeting

Additional Group Affiliations

2022–2025	Collaborator	JWST Titan Guaranteed Time Observation Team
2018–present	Collaborator	NASA Astrobio. Inst. Habitability of Hydrocarbon Worlds Group

Public Communication and Community Service

Scientific Consultant for NASA Science Visualization Studio

Invited Public Speaker for geology and astronomy at *Bedford Garden Club* (2025), *Ward Pound Ridge Reservation* (2024), *Pound Ridge Land Conservancy* (2023, 2024), *Trailside Museum Children’s Summer Ecology Program* (2023), *Trailside Nature Museum* (2022)

Judge at the *Westchester Junior Science and Humanities Symposium* (2024)

Overnight Near-to-Peer Mentor to group of 5 high school students at the GSFC *Boys Night In* (2019)

Volunteer and Exhibitor representing Cassini CIRS and Lunar Reconnaissance Orbiter (LRO); *International Observe the Moon Night* at GSFC (2018-2019); *Awesome-Con* at Washington, DC (2018-2019); *Celebrating Earth Day with NASA at Union Station* (2017-2019); *Space Night* with the Bowie Baysox minor league baseball team (2018); *US Science and Technology Expo* in Washington, DC (2019); *Science Jamboree* at GSFC (2017-2018); *Outer Space in the President’s Place* in the Eisenhower Executive Office Building in Washington, DC (2017)

Convener of the CCSU Physics Open House for local high school students (2015, 2016)

Panelist 22nd Annual Science and Technology Brain Trust, US News STEM Solutions Conference (2014)